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Course code: DSA0602

Course name: Data Handling and Visualization

LAB PROGRAMS (1-7)

SET 1: MONTHLY SALES DATA

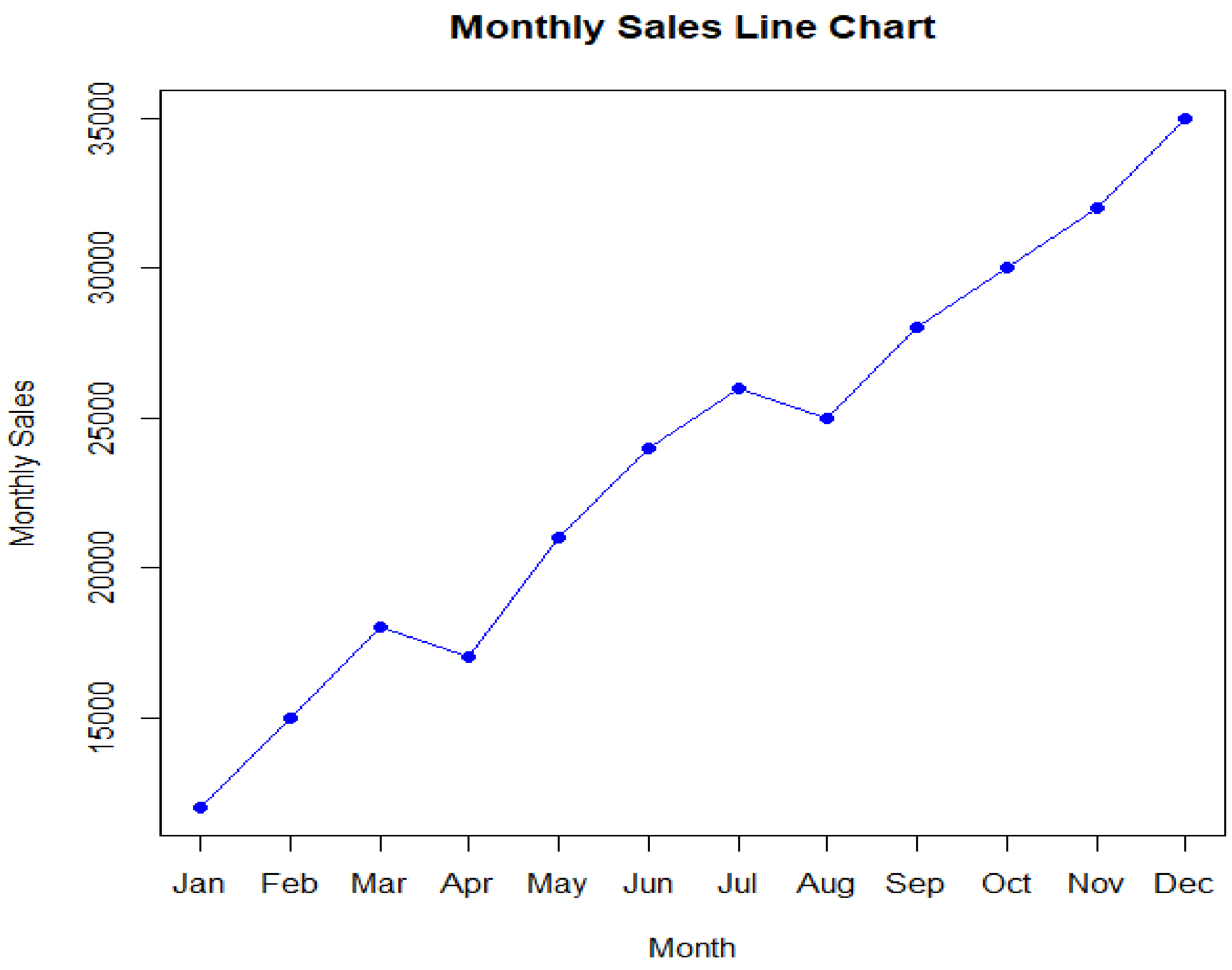
Dataset: Monthly Sales

```
# Monthly Sales Dataset

sales <- data.frame(

  Month = c("Jan","Feb","Mar","Apr","May","Jun",

           "Jul","Aug","Sep","Oct","Nov","Dec"),

  Sales = c(12000,15000,18000,17000,21000,24000,

           26000,25000,28000,30000,32000,35000),

  Advertising = c(2000,2500,3000,2800,3500,4000,

                4200,4100,4500,5000,5200,5500)

)

Sales
```

	Month	Sales	Advertising
1	Jan	12000	2000
2	Feb	15000	2500
3	Mar	18000	3000
4	Apr	17000	2800
5	May	21000	3500
6	Jun	24000	4000
7	Jul	26000	4200
8	Aug	25000	4100
9	Sep	28000	4500
10	Oct	30000	5000
11	Nov	32000	5200
12	Dec	35000	5500

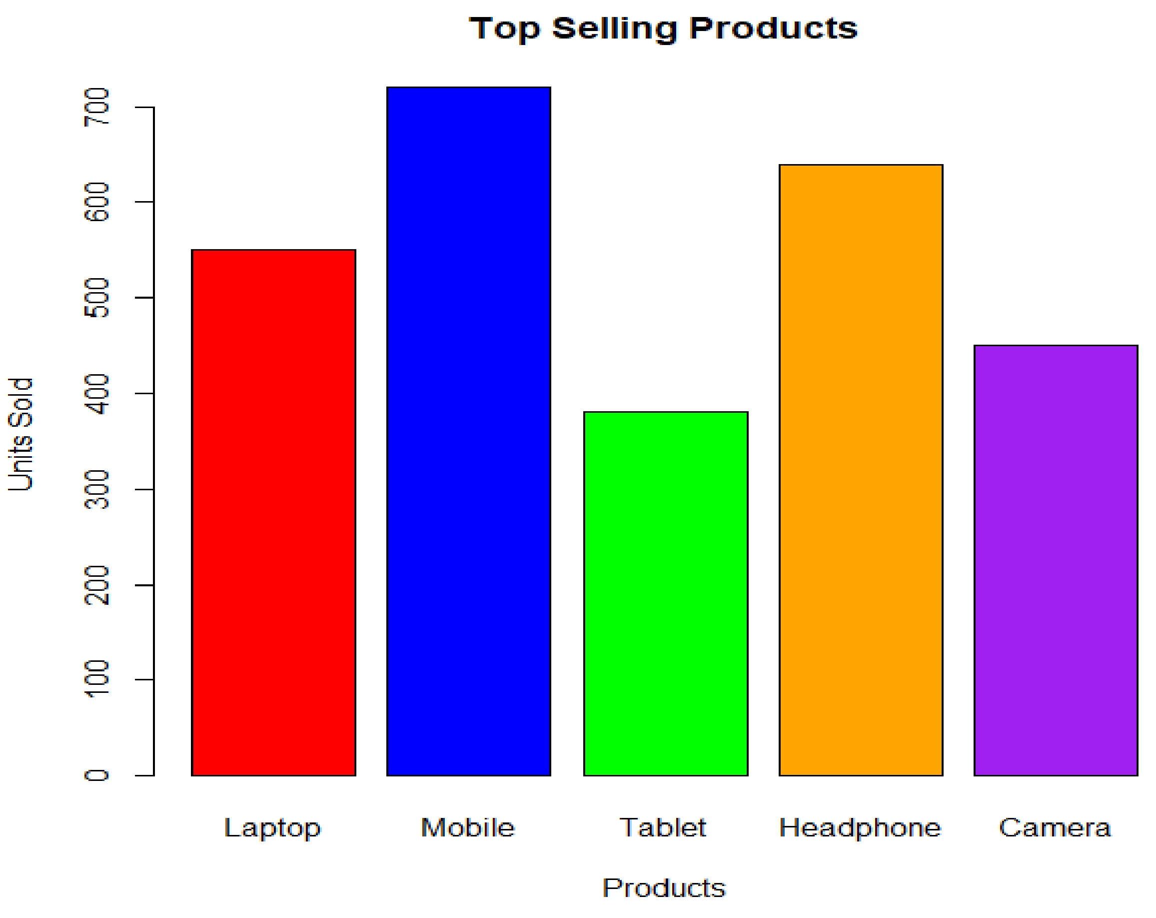
1. **Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.**

```
# Line Chart
plot(sales$Sales,
    type="o",
    col="blue",
    pch=16,
    xaxt="n",
    xlab="Month",
    ylab="Monthly Sales",
    main="Monthly Sales Line Chart")
axis(1,
    at=1:12,
    labels=sales$Month)
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

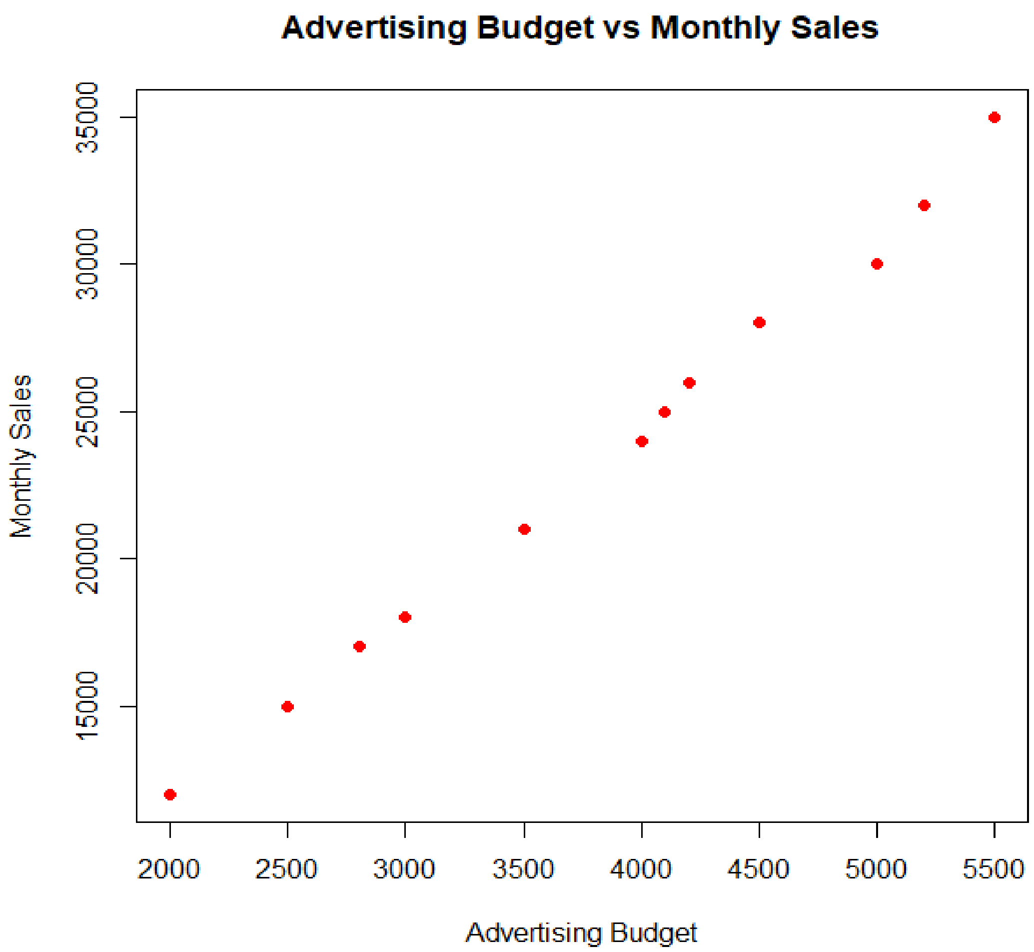
```
# Product Sales Datase

product <- data.frame(

 Product = c("Laptop","Mobile","Tablet","Headphone","Camera"),

 Sales = c(550,720,380,640,450)

)

# Colorful Bar Chart

barplot(product$Sales,

        names.arg = product$Product,

        col = c("red","blue","green","orange","purple"),

        xlab = "Products",

        ylab = "Units Sold",

        main = "Top Selling Products")
```



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

```
# Scatter Plot
plot(sales$Advertising,
    sales$Sales,
    pch=19,
    col="red",
    xlab="Advertising Budget",
    ylab="Monthly Sales",
    main="Advertising Budget vs Monthly Sales")
```

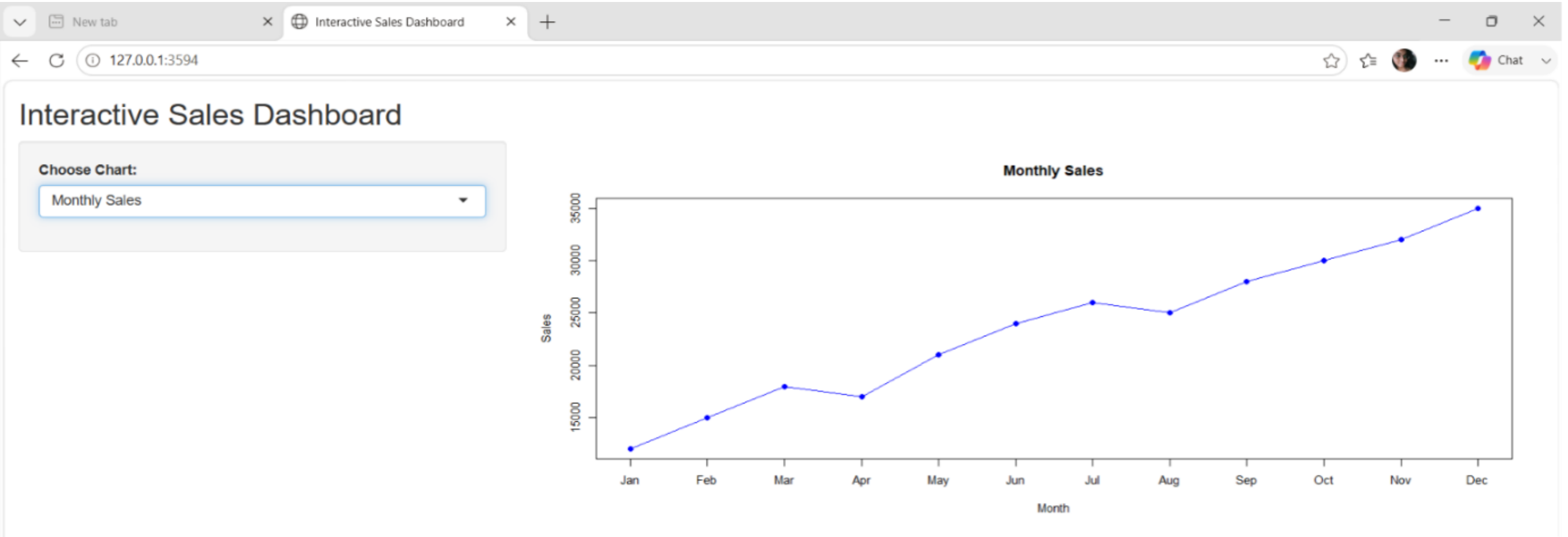


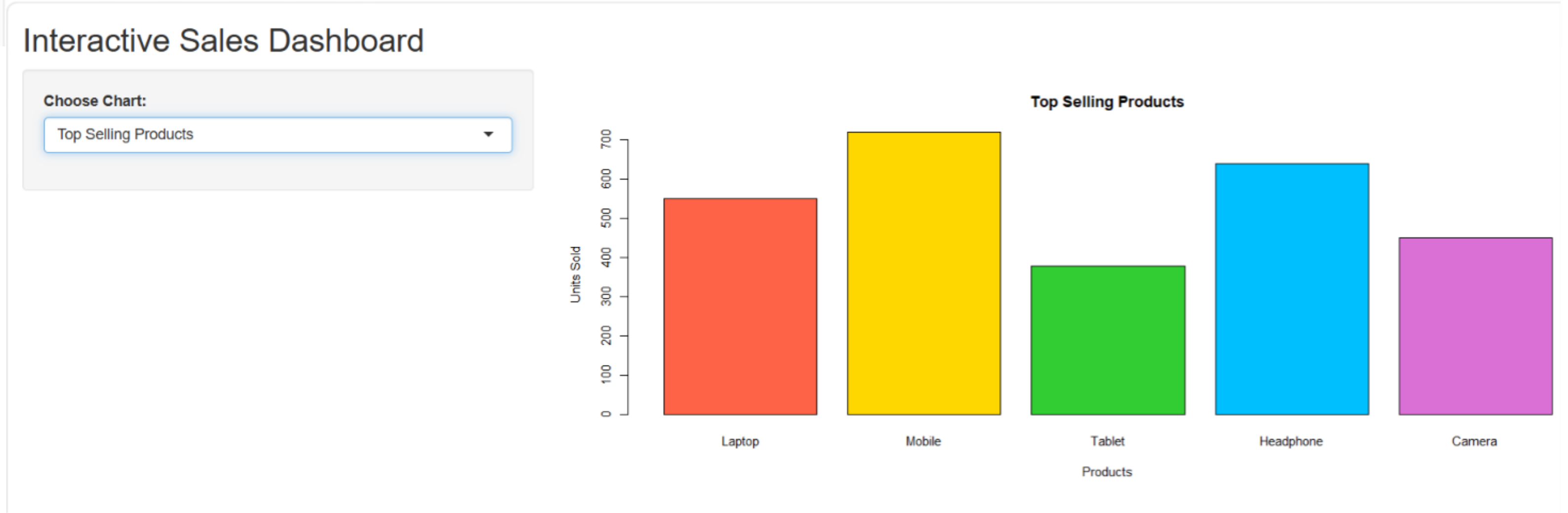
4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore

sales data interactively.

```
library(shiny)

# Monthly Sales Data
sales <- data.frame(
  Month = c("Jan","Feb","Mar","Apr","May","Jun",
            "Jul","Aug","Sep","Oct","Nov","Dec"),
  Sales = c(12000,15000,18000,17000,21000,24000,
            26000,25000,28000,30000,32000,35000)
)

# Product Sales Data
product <- data.frame(
  Product = c("Laptop","Mobile","Tablet","Headphone","Camera"),
  Sales = c(550,720,380,640,450)
)

# UI
ui <- fluidPage(

  titlePanel("Interactive Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("chart",
                  "Choose Chart:",
                  choices = c("Monthly Sales","Top Selling Products"))

    ),

    mainPanel(

      plotOutput("plot")

    )
```

```r
  )

)

# Server
server <- function(input, output){

  output$plot <- renderPlot({

    if(input$chart=="Monthly Sales"){

      plot(sales$Sales,
           type="o",
           col="blue",
           pch=16,
           xaxt="n",
           xlab="Month",
           ylab="Sales",
           main="Monthly Sales")

      axis(1, at=1:12, labels=sales$Month)

    }

    else{

      barplot(product$Sales,
              names.arg=product$Product,
              col=c("tomato","gold","limegreen","deepskyblue","orchid"),
              border="black",
              xlab="Products",
              ylab="Units Sold",
              main="Top Selling Products")

    }

  })

}

shinyApp(ui, server)
```





SET 2: CUSTOMER FEEDBACK ANALYSIS

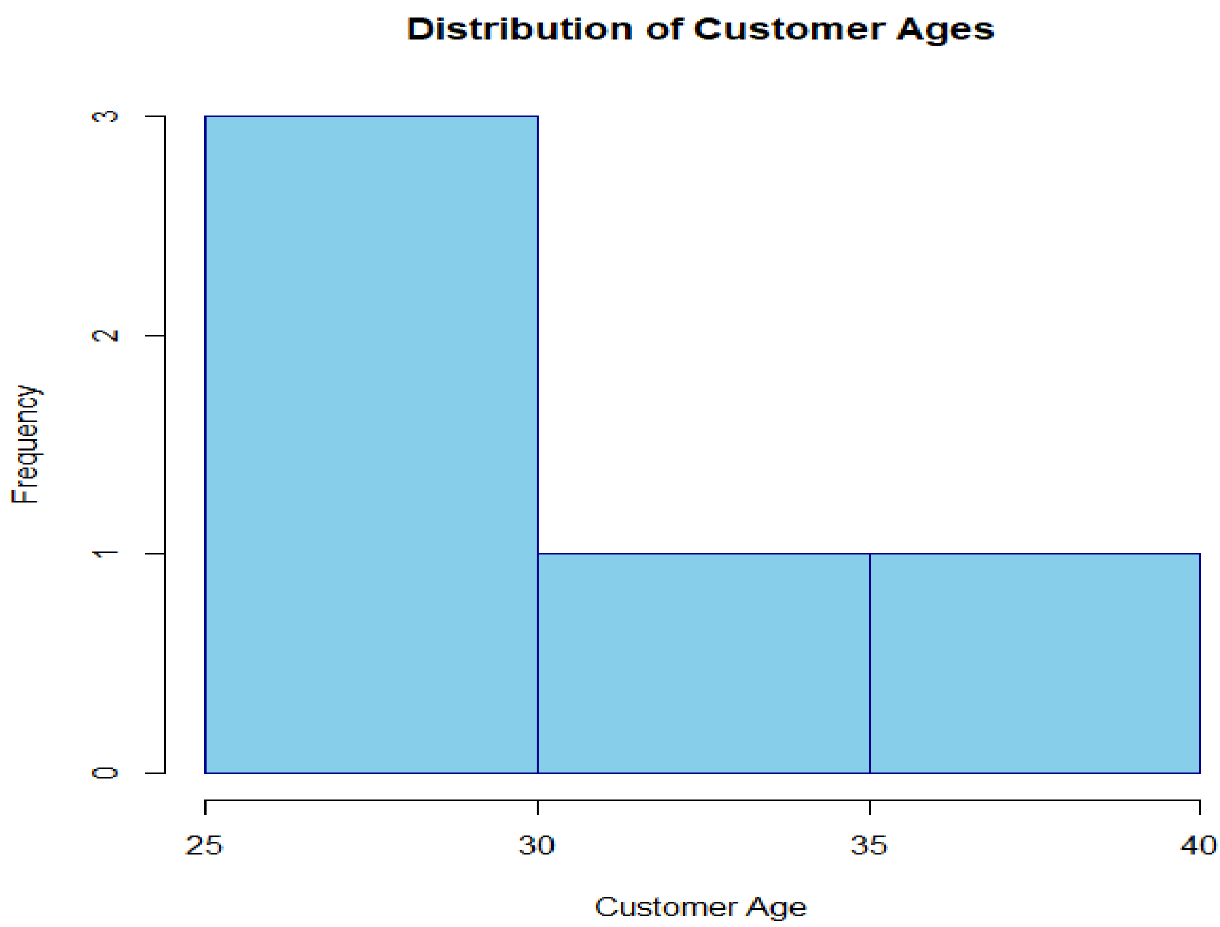
Dataset

```
# Customer Satisfaction Dataset
customer <- data.frame(
 CustomerID = c(1,2,3,4,5),
 Age = c(25,30,35,28,40),
 Satisfaction = c(4,5,3,4,5)
)
customer
```

```
customer
CustomerID Age Satisfaction
         1  25            4
         2  30            5
         3  35            3
         4  28            4
         5  40            5
```

1. **Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.**

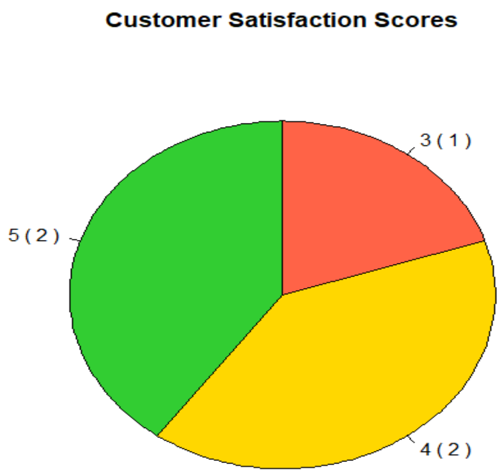
```
hist(customer$Age,
    breaks = 5,
    col = "skyblue",
    border = "darkblue",
    main = "Distribution of Customer Ages",
    xlab = "Customer Age",
    ylab = "Frequency")
```



2.Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

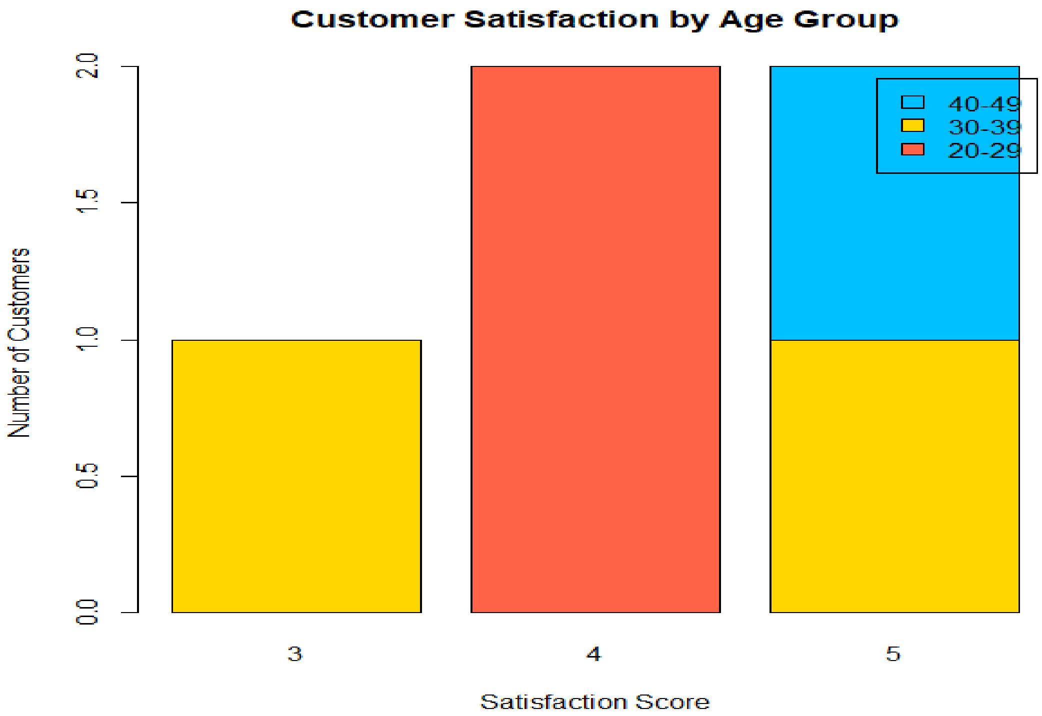
```
score <- table(customer$Satisfaction)

pie(score,

   labels = paste(names(score),"(",score,")"),

   col = c("tomato","gold","limegreen"),

   main = "Customer Satisfaction Scores",
```

clockwise =



3.Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
customer$AgeGroup <- c("20-29","30-39","30-39","20-29","40-49")
data <- table(customer$AgeGroup,
        customer$Satisfaction)
barplot(data,
    col = c("tomato","gold","deepskyblue"),
    border = "black",
    legend.text = rownames(data),
    xlab = "Satisfaction Score",
    ylab = "Number of Customers",
    main = "Customer Satisfaction by Age Group")
```



4.Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
library(tm)
library(wordcloud)
library(RColorBrewer)
feedback <- c(
"Good service",
"Excellent support",
"Very satisfied",
"Friendly staff",
"Excellent service"
)
wordcloud(feedback,
        min.freq = 1,
        random.order = FALSE,
        rot.per = 0.25,
        colors = brewer.pal(8,"Dark2"))
```

SET 3: EMPLOYEE PERFORMANCE EVALUATION

Dataset: Employee Performance

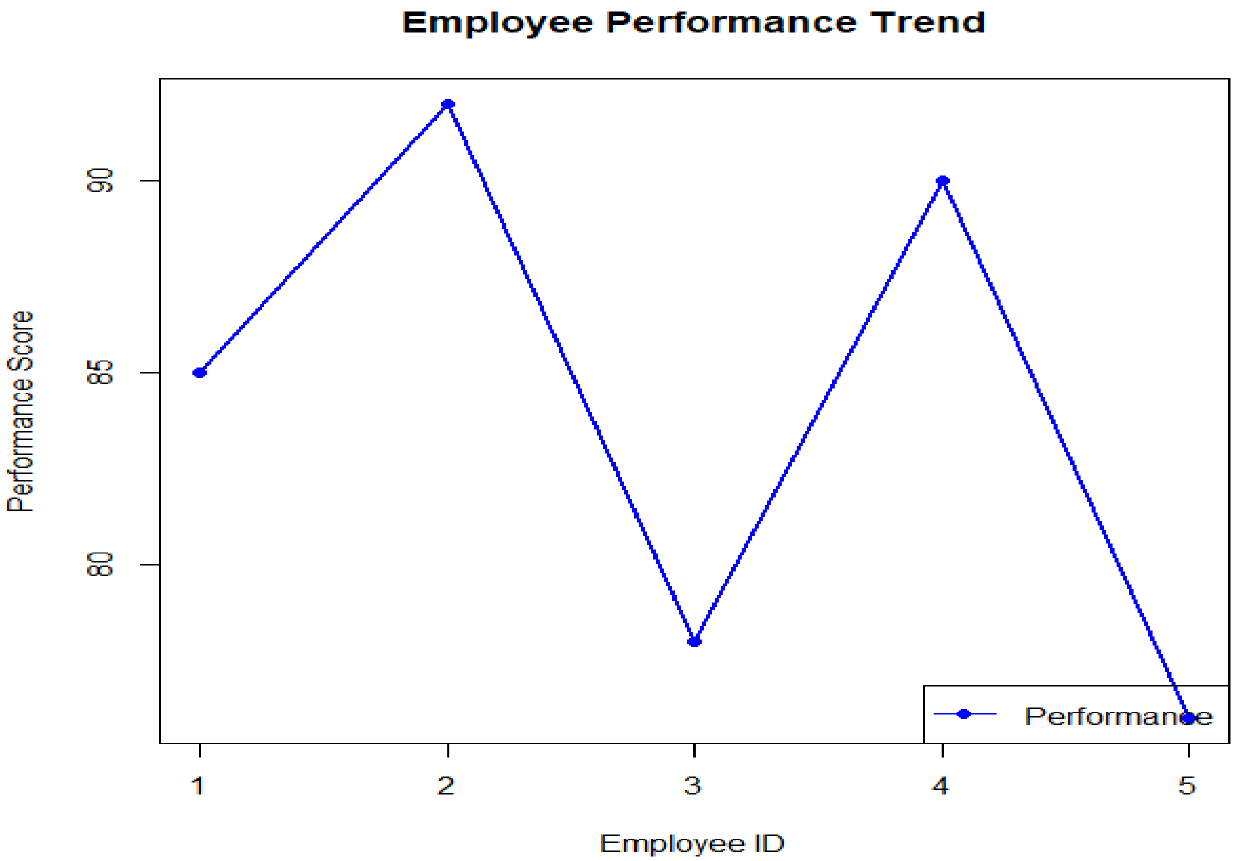
```
employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)
employee
```

```
employee
EmployeeID Department YearsService Performance
         1      Sales            5          85
         2         HR            3          92
         3  Marketing            7          78
         4      Sales            4          90
         5         HR            2          76
```

1. **Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.**

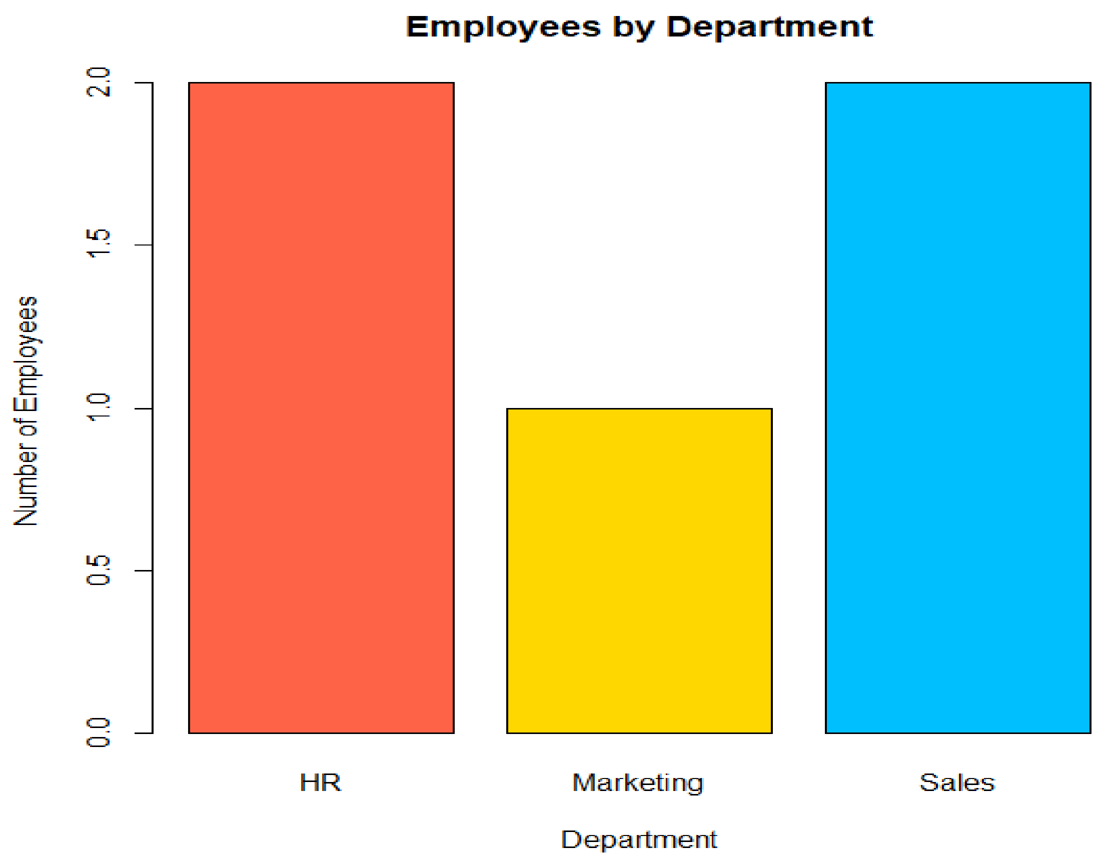
```
plot(employee$EmployeeID,
   employee$Performance,
   type = "o",
   pch = 16,
   col = "blue",
```

```
    lwd = 2,
    xlab = "Employee ID",
    ylab = "Performance Score",
    main = "Employee Performance Trend")

legend("bottomright",
       legend = "Performance",
       col = "blue",
       lty = 1,
       pch = 16)
```



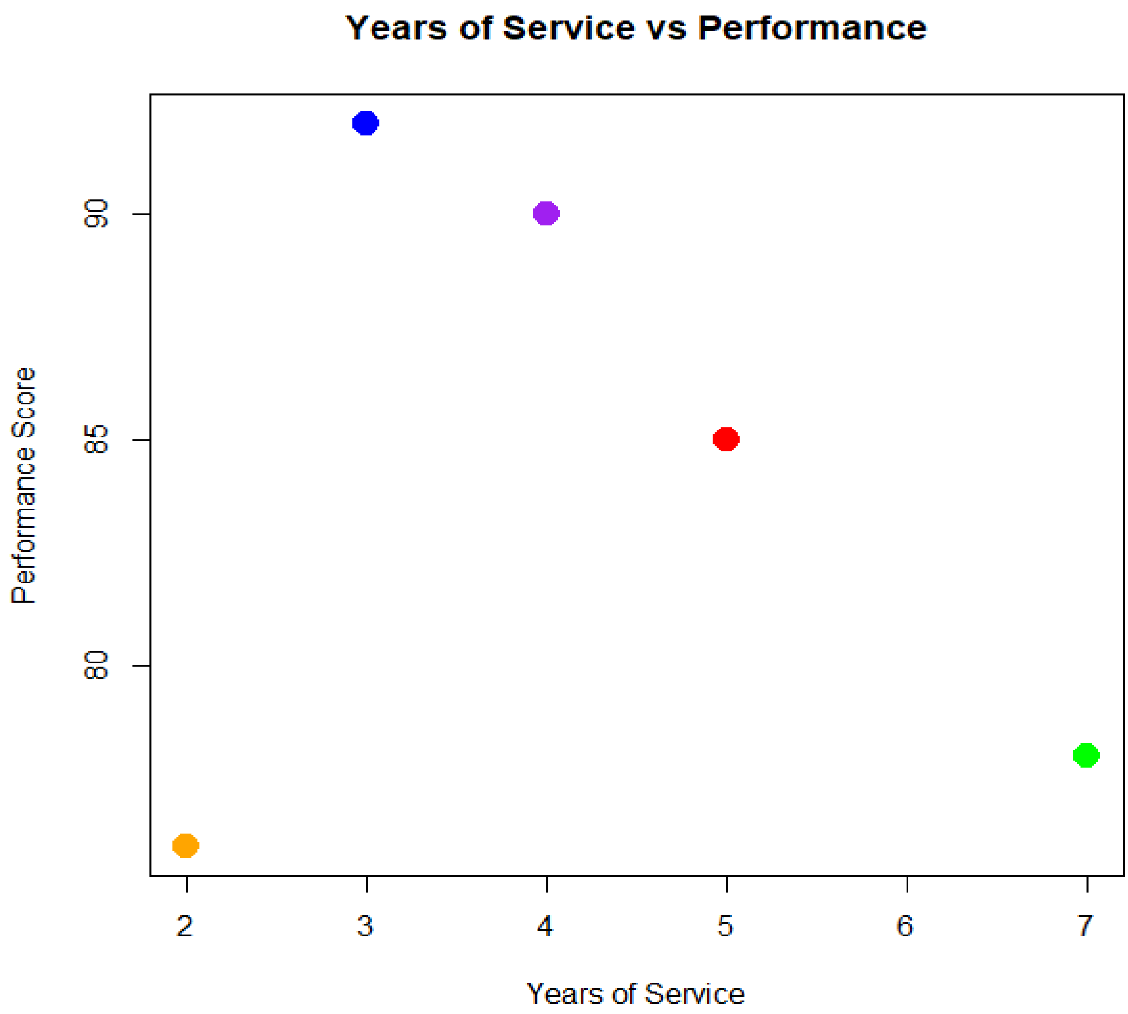
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
dept <- table(employee$Department)

barplot(dept,
        col = c("tomato","gold","deepskyblue"),
        border = "black",
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees")
```



3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
plot(employee$YearsService,
    employee$Performance,
    pch = 19,
    col = c("red","blue","green","purple","orange"),
    cex = 2,
    xlab = "Years of Service",
    ylab = "Performance Score",
    main = "Years of Service vs Performance")
```

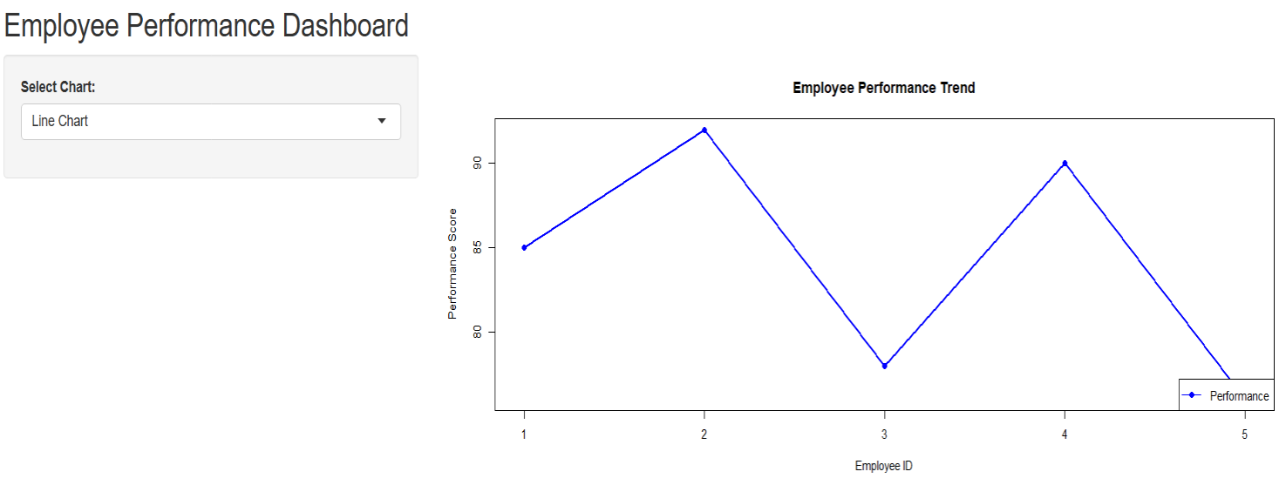


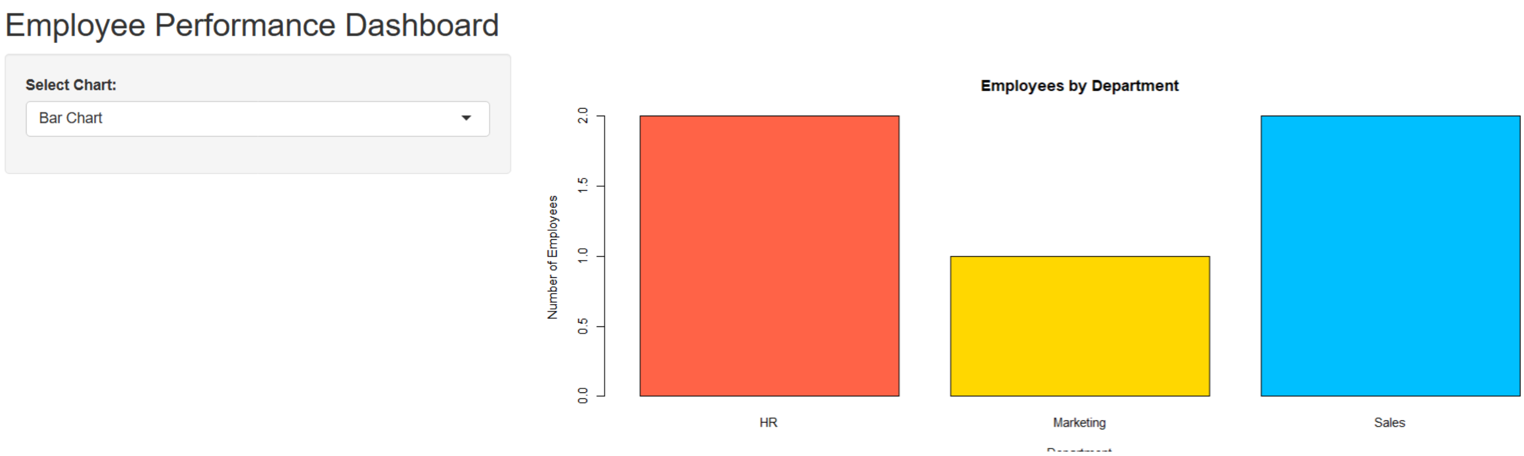
4. **Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.**

```
library(shiny)

# Product Inventory Dataset
product <- data.frame(
  ProductID = c(1,2,3,4,5),
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220),
  Category = c("Electronics","Electronics","Furniture","Furniture","Accessories"),
  Price = c(500,350,700,450,300)
)
# Create data for stacked bar chart
inventory <- table(product$Category, product$ProductName)
```

```r
# User Interface
ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Bar Chart",
                              "Stacked Bar Chart"))
    ),
    mainPanel(
      plotOutput("plot")
    )
  )
)
# Server
server <- function(input, output){
  output$plot <- renderPlot({
    if(input$chart == "Bar Chart"){
      barplot(product$Quantity,
              names.arg = product$ProductName,
              col = c("tomato","gold","limegreen","deepskyblue","orchid"),
              border = "black",
              xlab = "Products",
              ylab = "Quantity Available",
              main = "Product Inventory")
    }
    else{
      barplot(inventory,
              col = c("tomato","gold","deepskyblue"),
              border = "black",
              legend.text = rownames(inventory),
              xlab = "Products",
              ylab = "Count",
              main = "Products by Category")
    }
  })
}
# Run Dashboard
```

```
shinyApp(ui = ui, server = server)
```





SET 4: PRODUCT INVENTORY MANAGEMENT

Dataset: Product Inventory

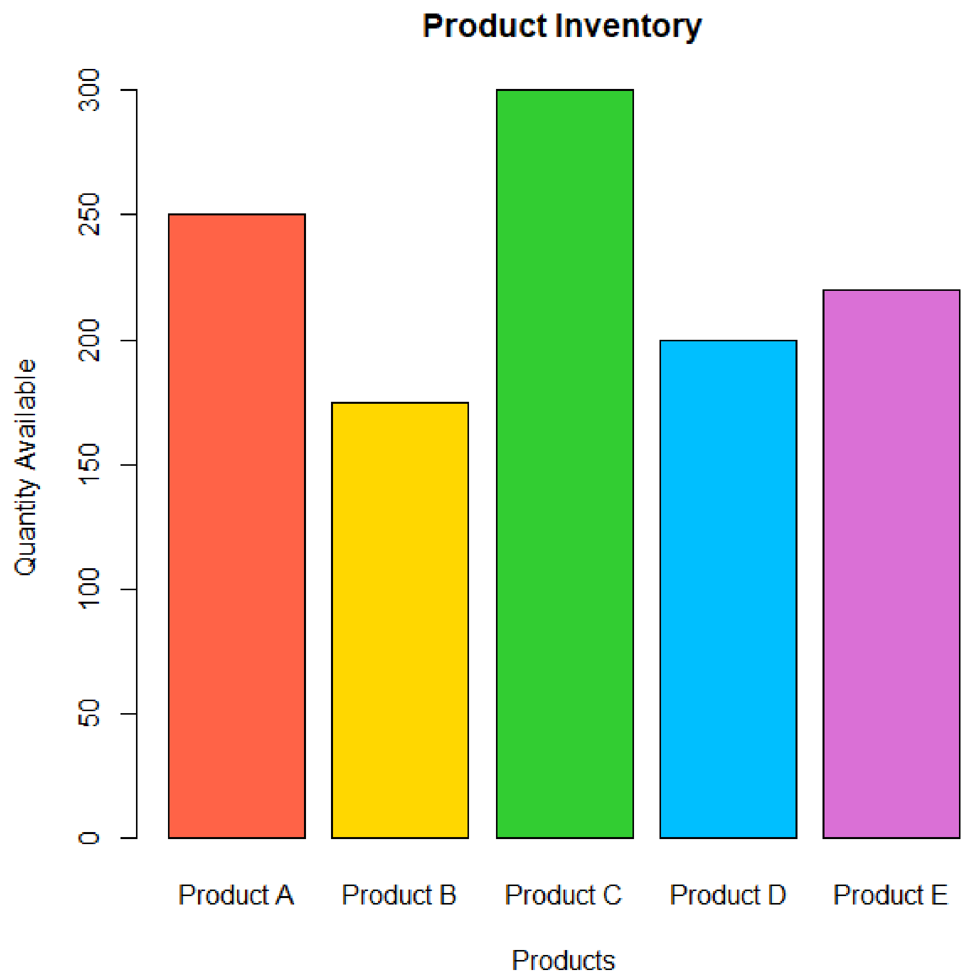
```
product <- data.frame(
  ProductID = c(1,2,3,4,5),
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220),
  Category = c("Electronics","Electronics","Furniture","Furniture","Accessories"),
  Price = c(500,350,700,450,300)
```

)Product

```
product
ProductID ProductName Quantity    Category Price
        1   Product A      250 Electronics   500
        2   Product B      175 Electronics   350
        3   Product C      300   Furniture   700
        4   Product D      200   Furniture   450
        5   Product E      220 Accessories   300
```

1. **Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.**

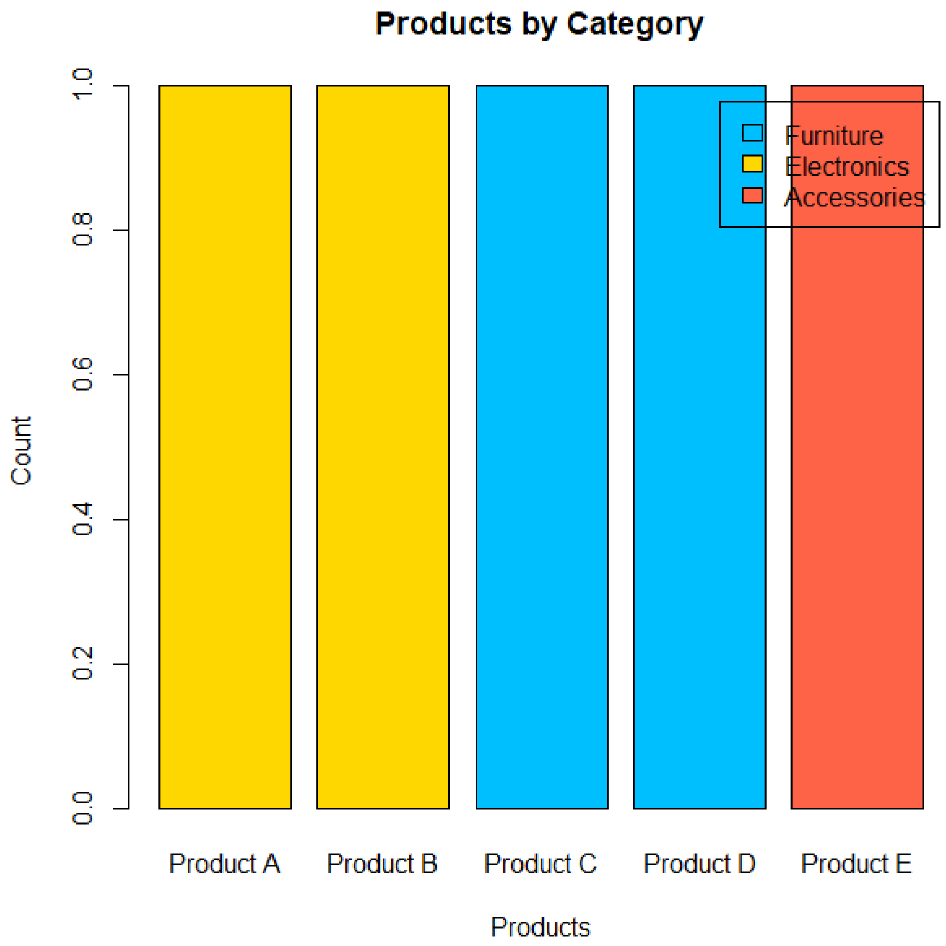
```
barplot(product$Quantity,
    names.arg = product$ProductName,
    col = c("tomato","gold","limegreen","deepskyblue","orchid"),
    border = "black",
    xlab = "Products",
    ylab = "Quantity Available",
    main = "Product Inventory")
```



2. **Generate a stacked bar chart to show the quantity of each product within different product categories.**

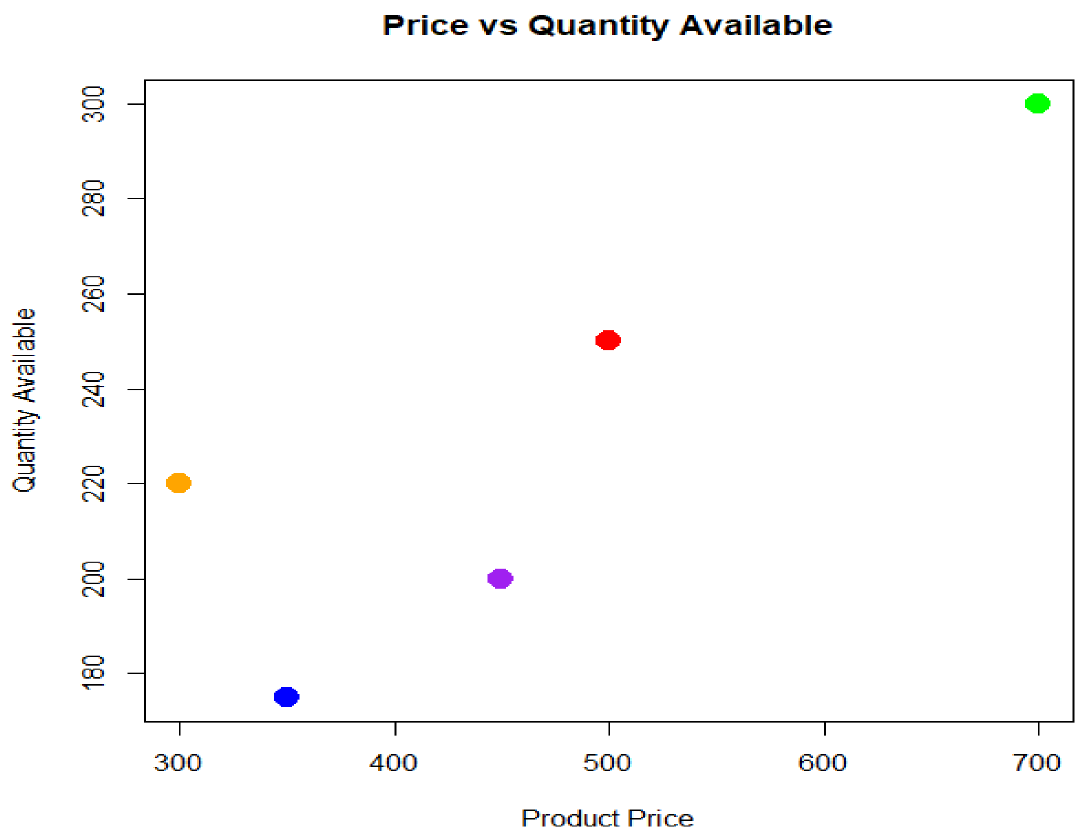
```
inventory <- table(product$Category,
              product$ProductName)

barplot(inventory,
      col = c("tomato","gold","deepskyblue"),
      border = "black",
      legend.text = rownames(inventory),
      xlab = "Products",
      ylab = "Count",
```

main = "Products by Category")

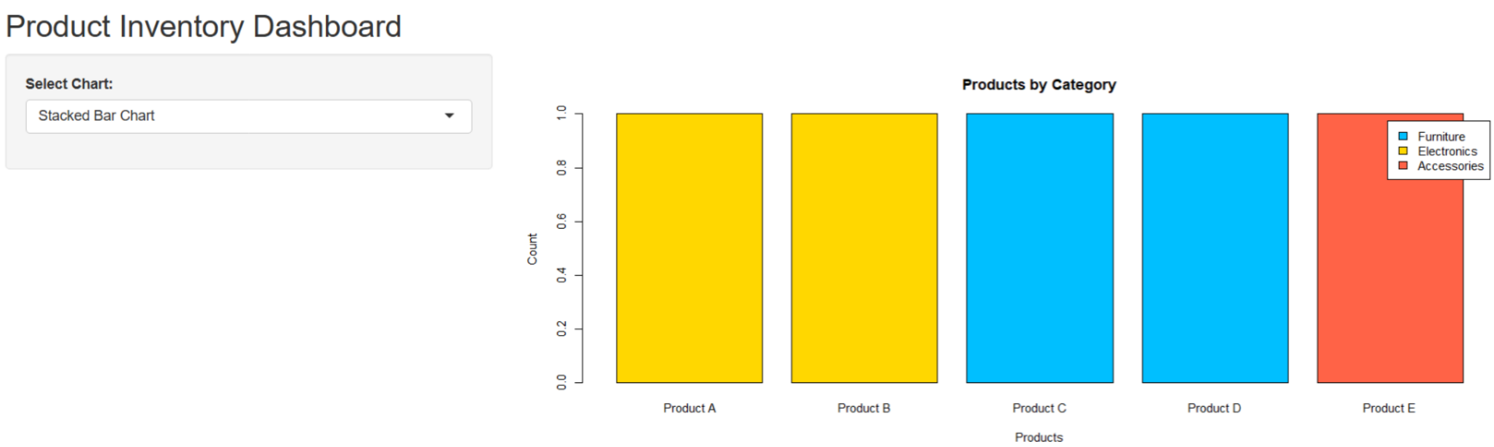


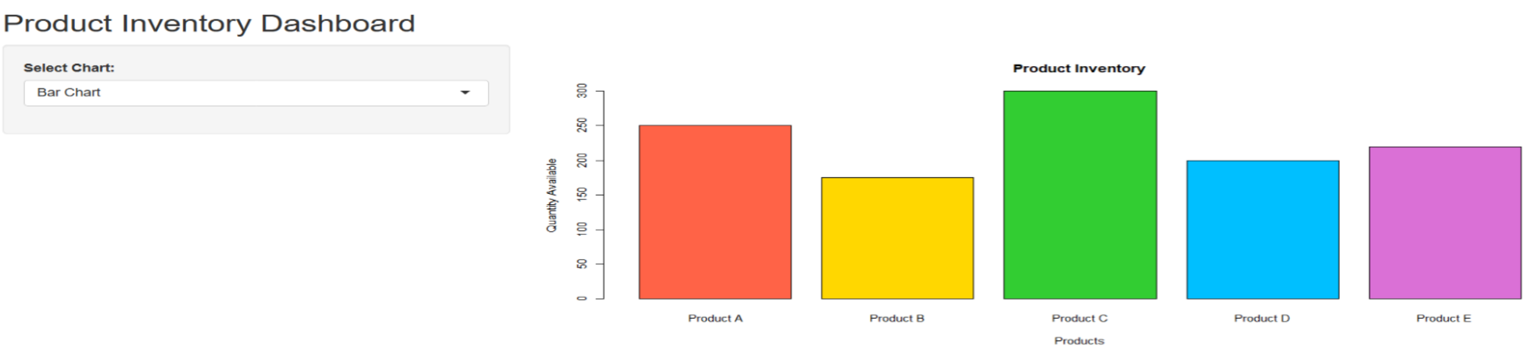
3. **Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.**

```
plot(product$Price,
   product$Quantity,
   pch = 19,
   col = c("red","blue","green","purple","orange"),
   cex = 2,
   xlab = "Product Price",
   ylab = "Quantity Available",
   main = "Price vs Quantity Available")
```



4. **Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.**





SET 5: WEBSITE ANALYTICS

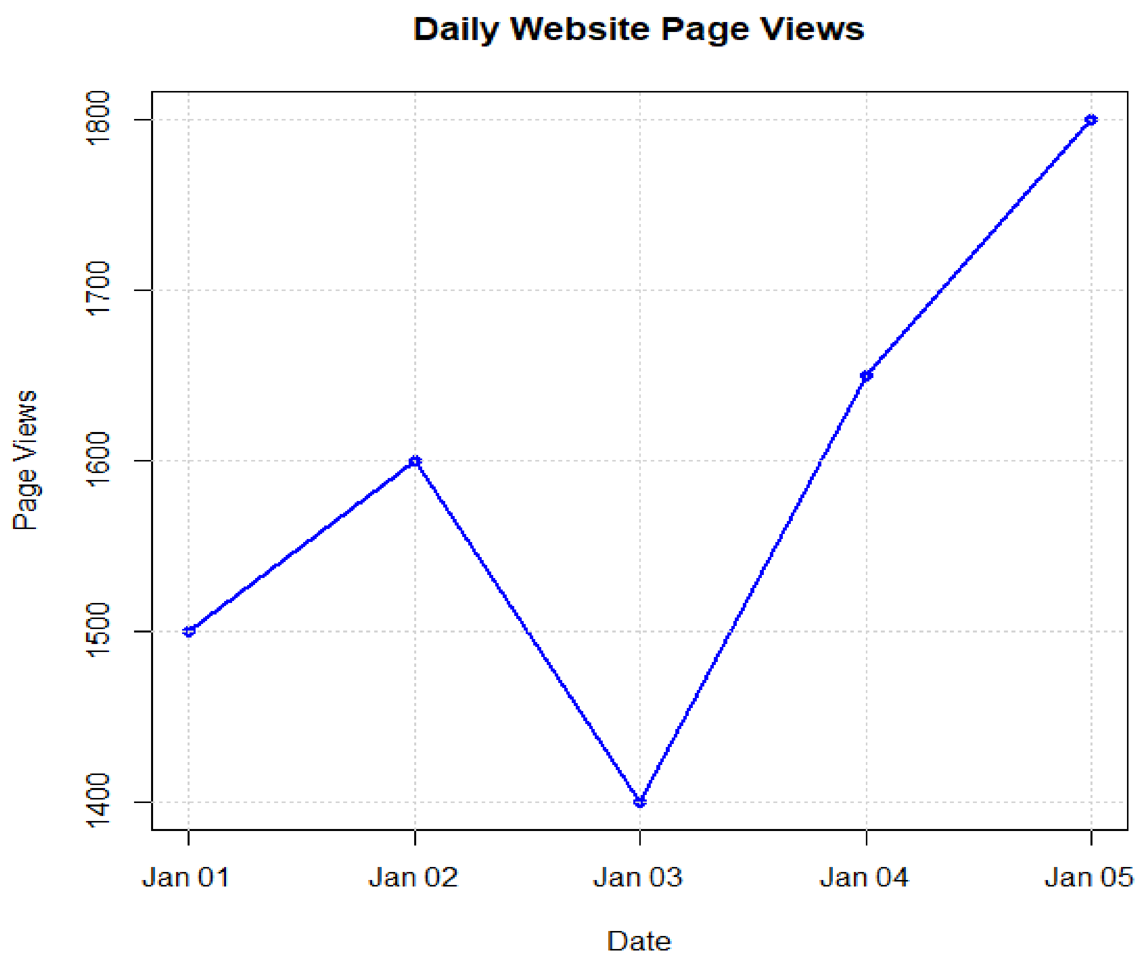
Dataset: Website Traffic

```
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03",
                   "2023-01-04","2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)
Traffic
```

```
        Date PageViews CTR
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
4 2023-01-04      1650 2.4
5 2023-01-05      1800 2.6
```

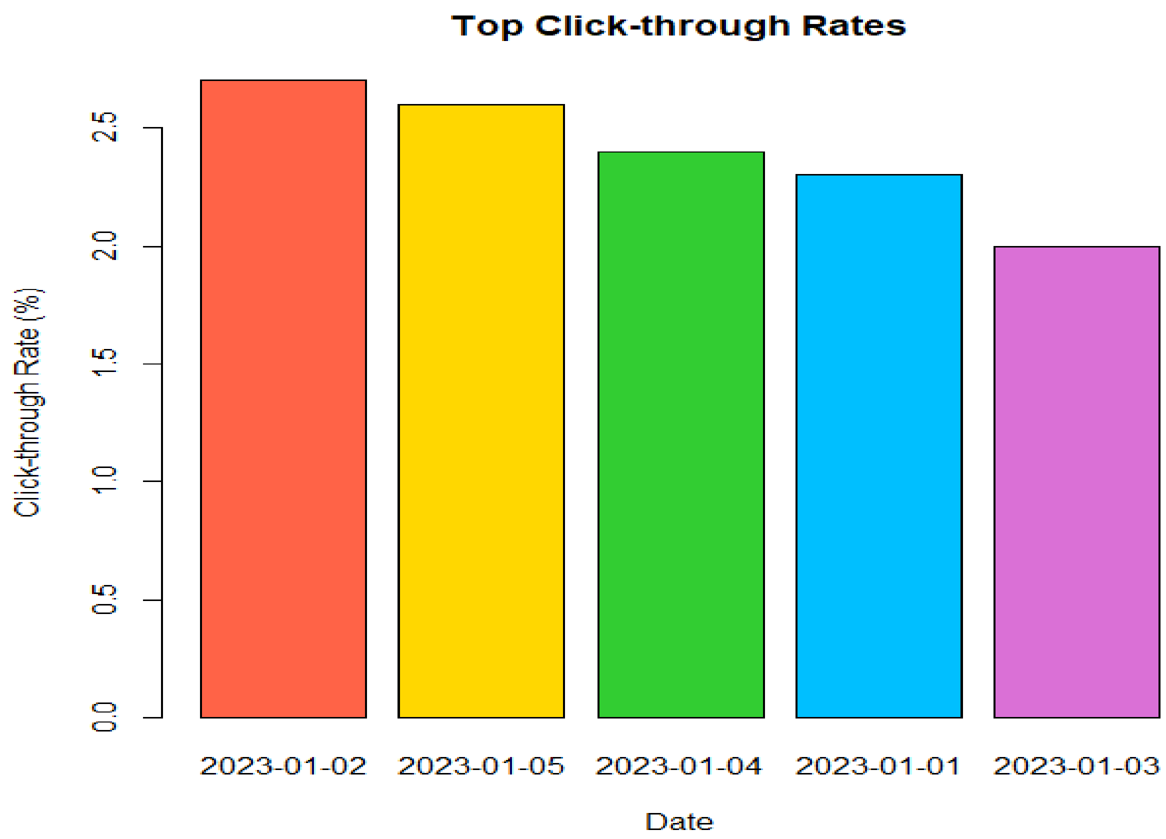
1. **Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.**

```
plot(traffic$Date,
   traffic$PageViews,
   type="o",
   pch=16,
   col="blue",
   lwd=2,
   xlab="Date",
   ylab="Page Views",
   main="Daily Website Page Views")
grid()
```



2. **Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.**

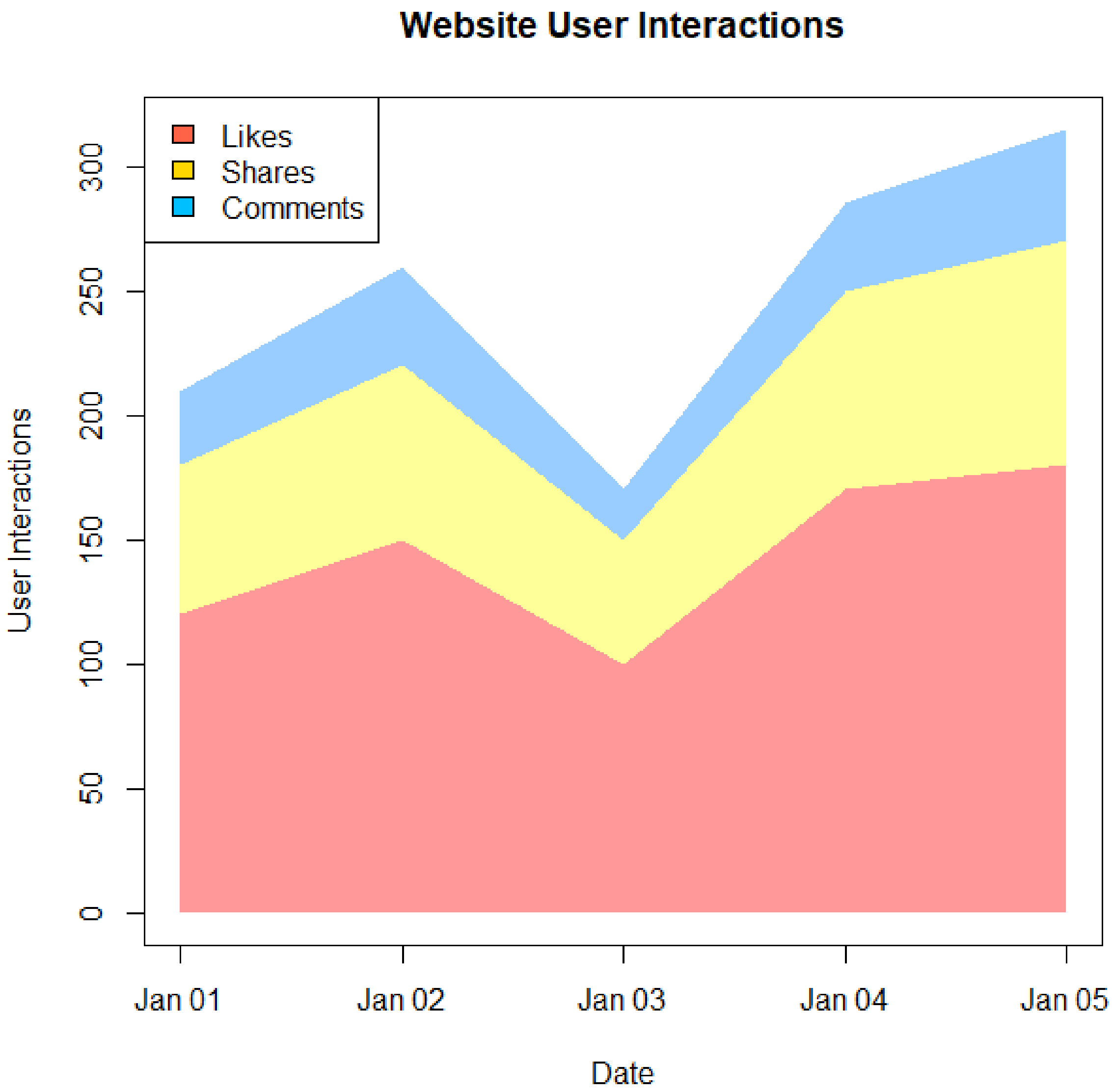
```
top <- traffic[order(-traffic$CTR),]

barplot(top$CTR,

    names.arg=top$Date,

    col=c("tomato","gold","limegreen","deepskyblue","orchid"),

    border="black",

    xlab="Date",

    ylab="Click-through Rate (%)",

    main="Top Click-through Rates")
```



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```
traffic$Likes <- c(120,150,100,170,180)

traffic$Shares <- c(60,70,50,80,90)

traffic$Comments <- c(30,40,20,35,45)

interaction <- rbind(

traffic$Likes,

traffic$Shares,

traffic$Comments)


col <- c("tomato","gold","deepskyblue")


plot(traffic$Date,

    traffic$Likes,

    type="n",
```

```
    ylim=c(0,max(colSums(interaction))),
    xlab="Date",
    ylab="User Interactions",
    main="Website User Interactions")

polygon(c(traffic$Date,rev(traffic$Date)),
        c(rep(0,5),rev(traffic$Likes)),
        col=rgb(1,0,0,0.4),
        border=NA)

polygon(c(traffic$Date,rev(traffic$Date)),
        c(traffic$Likes,
          rev(traffic$Likes+traffic$Shares)),
        col=rgb(1,1,0,0.4),
        border=NA)

polygon(c(traffic$Date,rev(traffic$Date)),
        c(traffic$Likes+traffic$Shares,
          rev(traffic$Likes+traffic$Shares+traffic$Comments)),
        col=rgb(0,0.5,1,0.4),
        border=NA)

legend("topleft",
       legend=c("Likes","Shares","Comments"),
       fill=col)
```

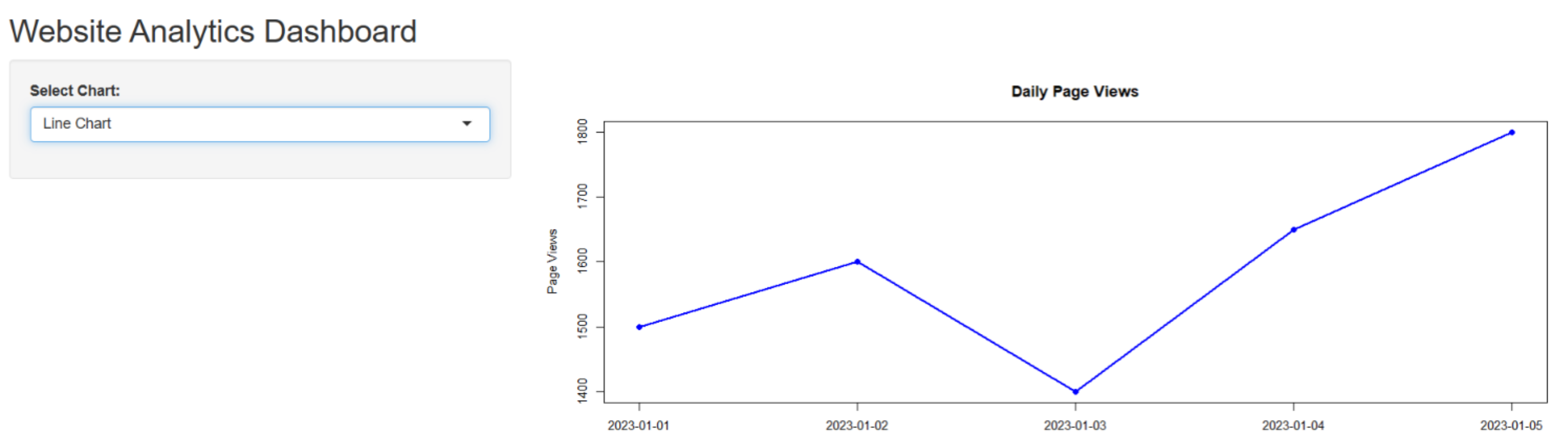


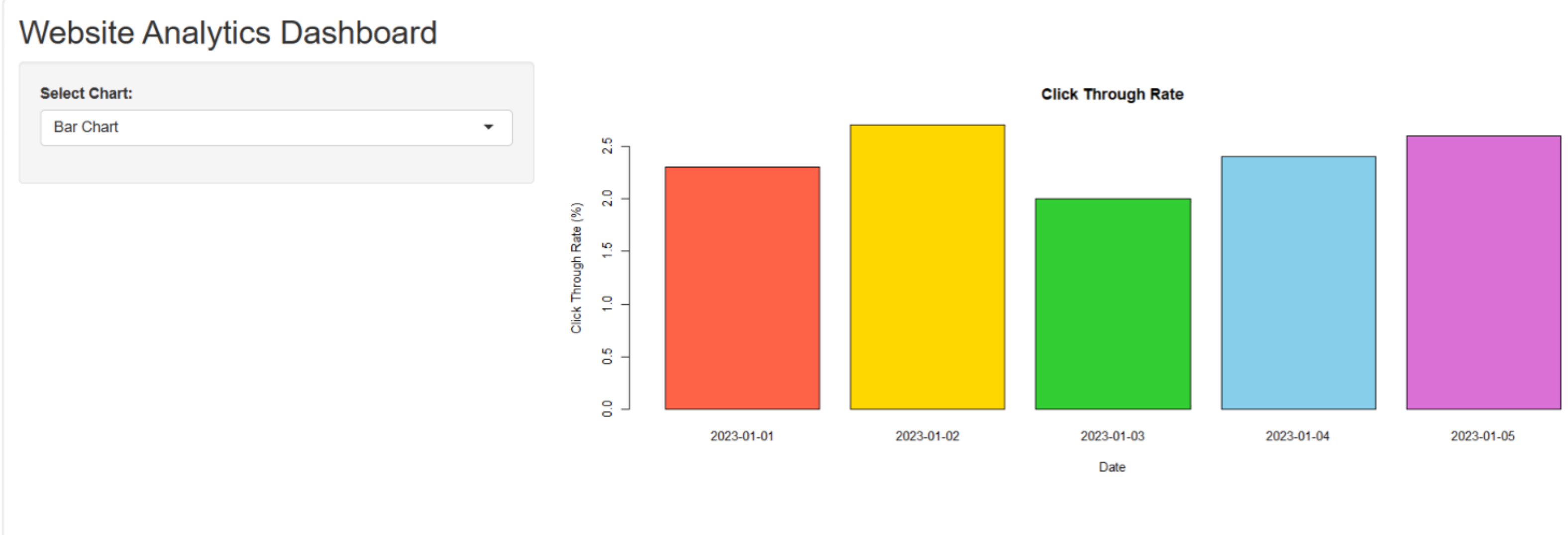
4.Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

```
library(shiny)
# Website Traffic Dataset
website <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05"),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)

# User Interface
ui <- fluidPage(
  titlePanel("Website Analytics Dashboard"),
```

```
  sidebarLayout(
    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Line Chart",
                              "Bar Chart"))
    ),

    mainPanel(
      plotOutput("plot")
    )
  )
)

# Server
server <- function(input, output){

  output$plot <- renderPlot({

    if(input$chart == "Line Chart"){

      plot(website$PageViews,
           type="o",
           col="blue",
           pch=16,
           lwd=2,
           xaxt="n",
           xlab="Date",
           ylab="Page Views",
           main="Daily Page Views")

      axis(1,
           at=1:5,
           labels=website$Date)

    }

    else{

      barplot(website$CTR,
              names.arg=website$Date,
```

```
        col=c("tomato","gold","limegreen","skyblue","orchid"),
        border="black",
        xlab="Date",
        ylab="Click Through Rate (%)",
        main="Click Through Rate")
  }

 })

}

# Run Dashboard
shinyApp(ui=ui, server=server)
```





SET 6 PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

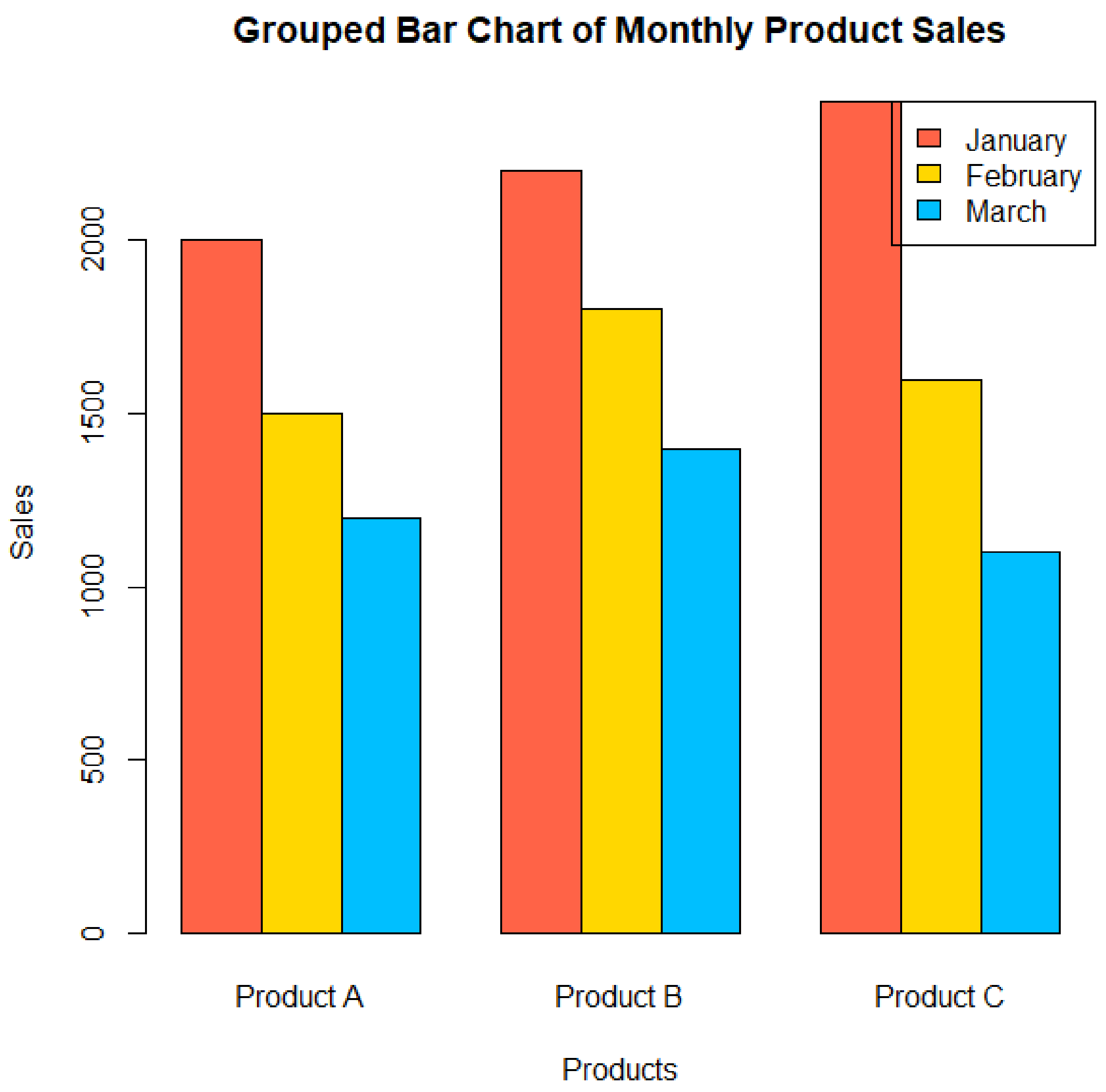
```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
Sales
```

```
> sales
    Product January February March
1 Product A    2000     2200  2400
2 Product B    1500     1800  1600
3 Product C    1200     1400  1100
>
```

1. **Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.**

```
# Dataset
sales <- data.frame(
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

# Create matrix
sales_matrix <- as.matrix(sales[,2:4])

# Grouped Bar Chart
barplot(sales_matrix,
        beside=TRUE,
        col=c("tomato","gold","deepskyblue"),
        names.arg=sales$ProductName,
        xlab="Products",
        ylab="Sales",
```

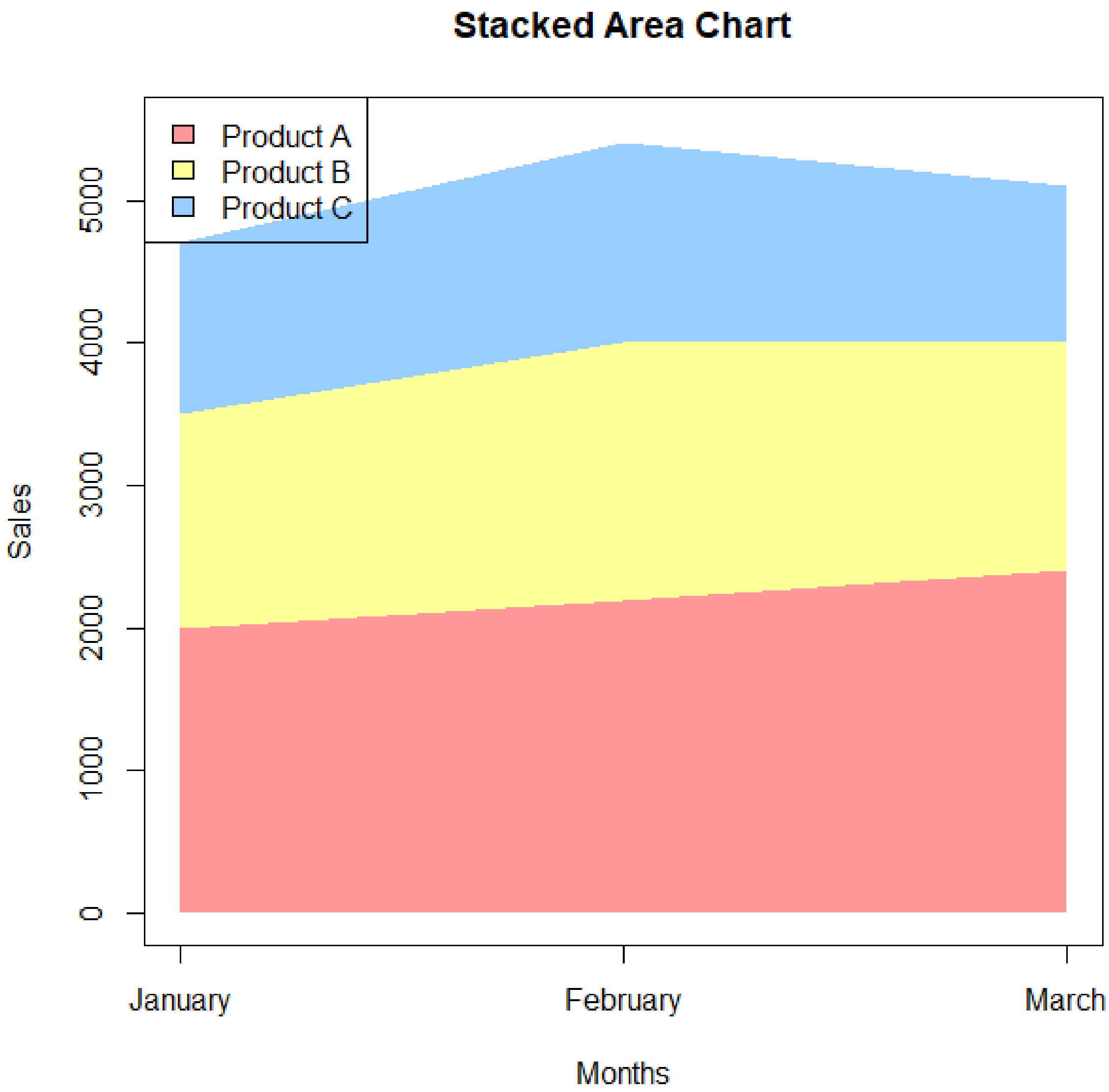
```
       main="Grouped Bar Chart of Monthly Product Sales")

legend("topright",
       legend=c("January","February","March"),
       fill=c("tomato","gold","deepskyblue"))
```



2. **Generate a stacked area chart to represent the overall sales trend for all products over the three months.**

```
months <- c(1,2,3)

ProductA <- c(2000,2200,2400)
ProductB <- c(1500,1800,1600)
ProductC <- c(1200,1400,1100)

plot(months,
```

```
    ProductA,
    type="n",
    ylim=c(0,5500),
    xaxt="n",
    xlab="Months",
    ylab="Sales",
    main="Stacked Area Chart")

axis(1,
    at=months,
    labels=c("January","February","March"))

polygon(c(months,rev(months)),
       c(rep(0,3),rev(ProductA)),
       col=rgb(1,0,0,0.4),
       border=NA)

polygon(c(months,rev(months)),
       c(ProductA,
         rev(ProductA+ProductB)),
       col=rgb(1,1,0,0.4),
       border=NA)

polygon(c(months,rev(months)),
       c(ProductA+ProductB,
         rev(ProductA+ProductB+ProductC)),
       col=rgb(0,0.5,1,0.4),
       border=NA)

legend("topleft",
      legend=c("Product A","Product B","Product C"),
      fill=c(rgb(1,0,0,0.4),
            rgb(1,1,0,0.4),
            rgb(0,0.5,1,0.4)))
```



3. **Build a table to show the monthly sales data for each product. Label the table elements.**

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

sales$TotalSales <- sales$January + sales$February + sales$March
# Sort by Total Sales (Highest to Lowest)
```

```
sales <- sales[order(-sales$TotalSales), ]

print(sales)
```

```
>
> print(sales)
  ProductID ProductName January February March TotalSales
1         1   Product A    2000     2200  2400       6600
2         2   Product B    1500     1800  1600       4900
3         3   Product C    1200     1400  1100       3700
>
```

4. **Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.**

```
library(shiny)

# Product Sales Dataset
sales <- data.frame(
 ProductID = c(1,2,3),
 ProductName = c("Product A","Product B","Product C"),
 January = c(2000,1500,1200),
 February = c(2200,1800,1400),
 March = c(2400,1600,1100)
)

# Sales Matrix
sales_matrix <- as.matrix(sales[,3:5])

ui <- fluidPage(

 titlePanel("Product Sales Analysis Dashboard"),

 sidebarLayout(

  sidebarPanel(

   selectInput("chart",
         "Select View",
         choices = c("Grouped Bar Chart",
               "Stacked Area Chart",
               "Sales Table"))

  ),
```

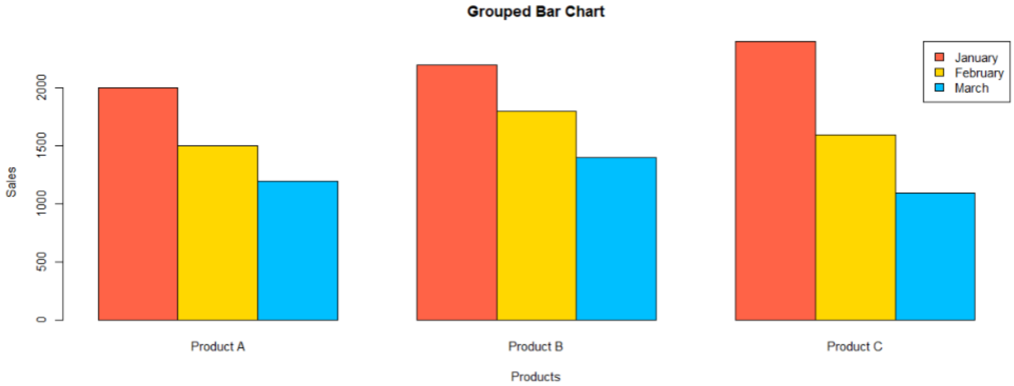
```
    mainPanel(

      plotOutput("plot"),

      tableOutput("table")

    )

  )

)

server <- function(input, output){

  output$plot <- renderPlot({

    if(input$chart == "Grouped Bar Chart"){

      barplot(sales_matrix,
              beside = TRUE,
              col = c("tomato","gold","deepskyblue"),
              names.arg = sales$ProductName,
              xlab = "Products",
              ylab = "Sales",
              main = "Grouped Bar Chart")

      legend("topright",
             legend = c("January","February","March"),
             fill = c("tomato","gold","deepskyblue"))

    }

    else if(input$chart == "Stacked Area Chart"){

      months <- c(1,2,3)

      A <- c(2000,2200,2400)
      B <- c(1500,1800,1600)
      C <- c(1200,1400,1100)
```

```
    plot(months,
         A,
         type="n",
         ylim=c(0,5500),
         xaxt="n",
         xlab="Months",
         ylab="Sales",
         main="Stacked Area Chart")

    axis(1,
         at=months,
         labels=c("January","February","March"))

    polygon(c(months,rev(months)),
            c(rep(0,3),rev(A)),
            col=rgb(1,0,0,0.4),
            border=NA)
    polygon(c(months,rev(months)),
            c(A,rev(A+B)),
            col=rgb(1,1,0,0.4),
            border=NA
    polygon(c(months,rev(months)),
            c(A+B,rev(A+B+C)),
            col=rgb(0,0.5,1,0.4),
            border=NA)
    legend("topleft",
           legend=c("Product A","Product B","Product C"),
           fill=c(rgb(1,0,0,0.4),
                  rgb(1,1,0,0.4),
                  rgb(0,0.5,1,0.4)))
  }
 })
 output$table <- renderTable({
  if(input$chart == "Sales Table"){

    sales
  }
 })
}
shinyApp(ui = ui, server = server)
```

Product Sales Analysis Dashboard

Select View

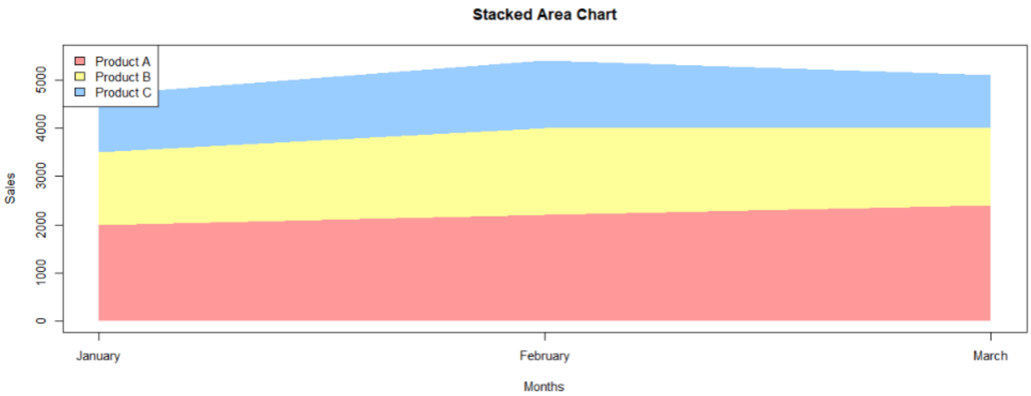
Grouped Bar Chart



Product Sales Analysis Dashboard

Select View

Stacked Area Chart



Product Sales Analysis Dashboard

Select View

Sales Table

ProductID	ProductName	January	February	March
1.00	Product A	2000.00	2200.00	2400.00
2.00	Product B	1500.00	1800.00	1600.00
3.00	Product C	1200.00	1400.00	1100.00

SET 7 CUSTOMER DEMOGRAPHICS ANALYSIS

Dataset: Customer Demographics

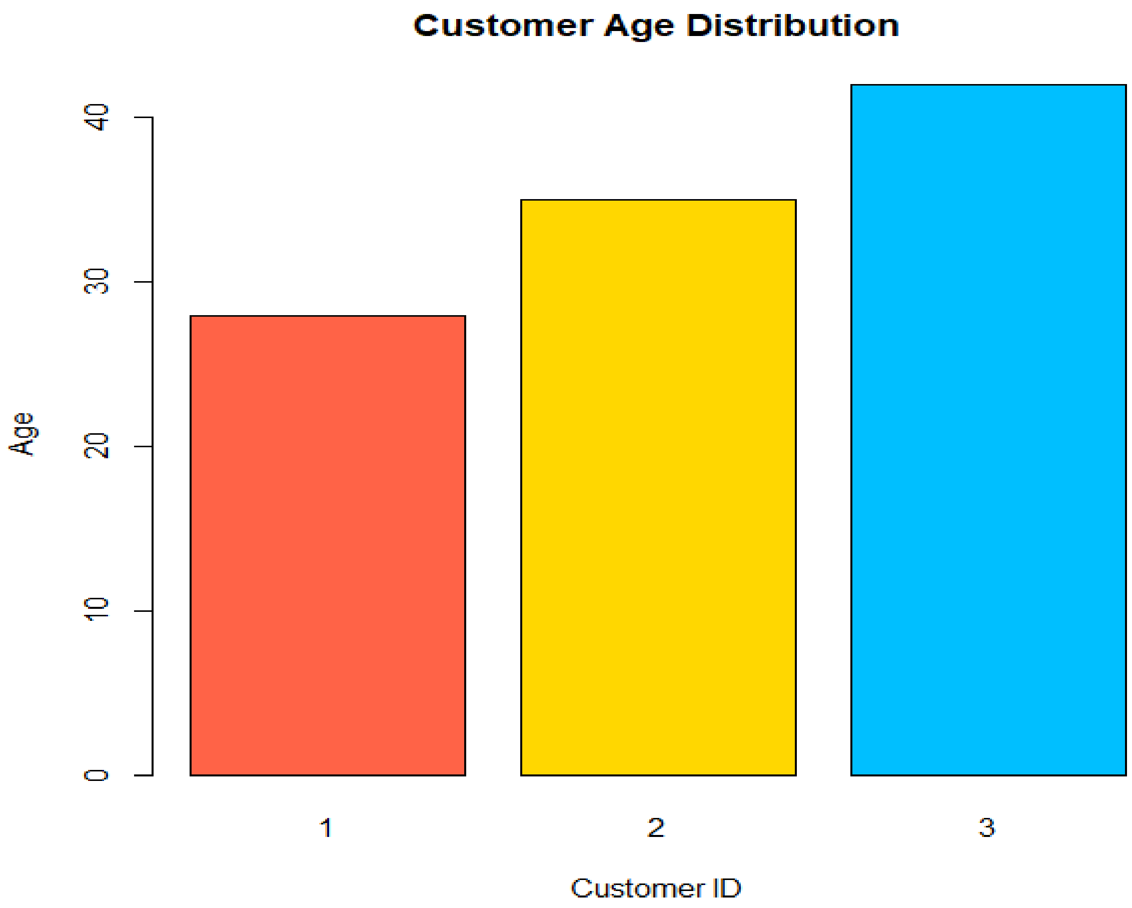
```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
```

Customer

```
> customer
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35   Male  60000
3          3  42 Female  75000
>
```

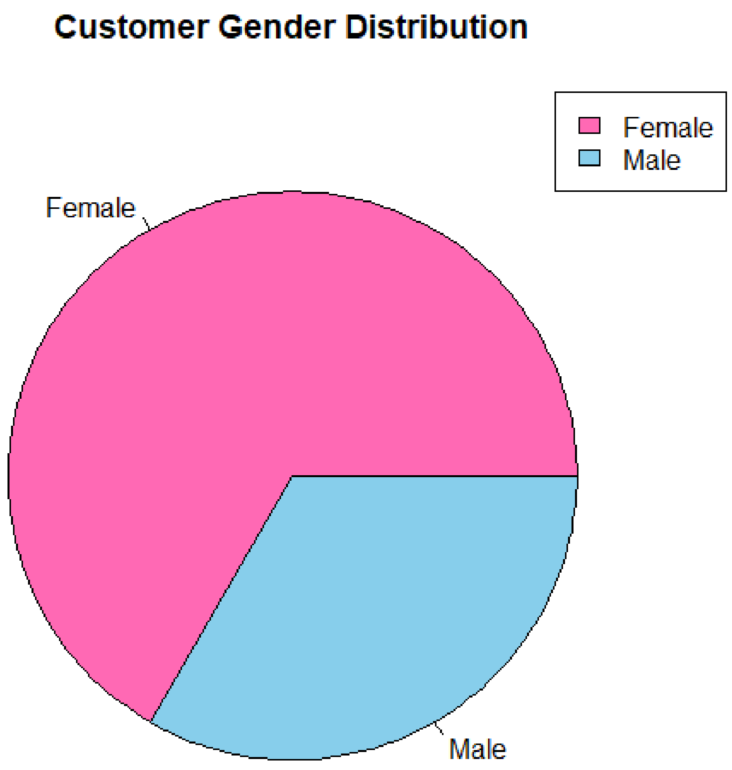
1. **Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.**

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

barplot(customer$Age,
        names.arg = customer$CustomerID,
        col = c("tomato","gold","deepskyblue"),
        border = "black",
        xlab = "Customer ID",
        ylab = "Age",
        main = "Customer Age Distribution")
```



2. Generate a pie chart to represent the distribution of customers by gender.

```
gender <- table(customer$Gender)

pie(gender,
  col = c("hotpink","skyblue"),
  main = "Customer Gender Distribution")

legend("topright",
     legend = names(gender),
     fill = c("hotpink","skyblue"))
```



3. **Build a table to show the demographic information for each customer. Label the table elements.**

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

customer$AgeGroup <- c("20-30","31-40","41-50")

print(customer)
```

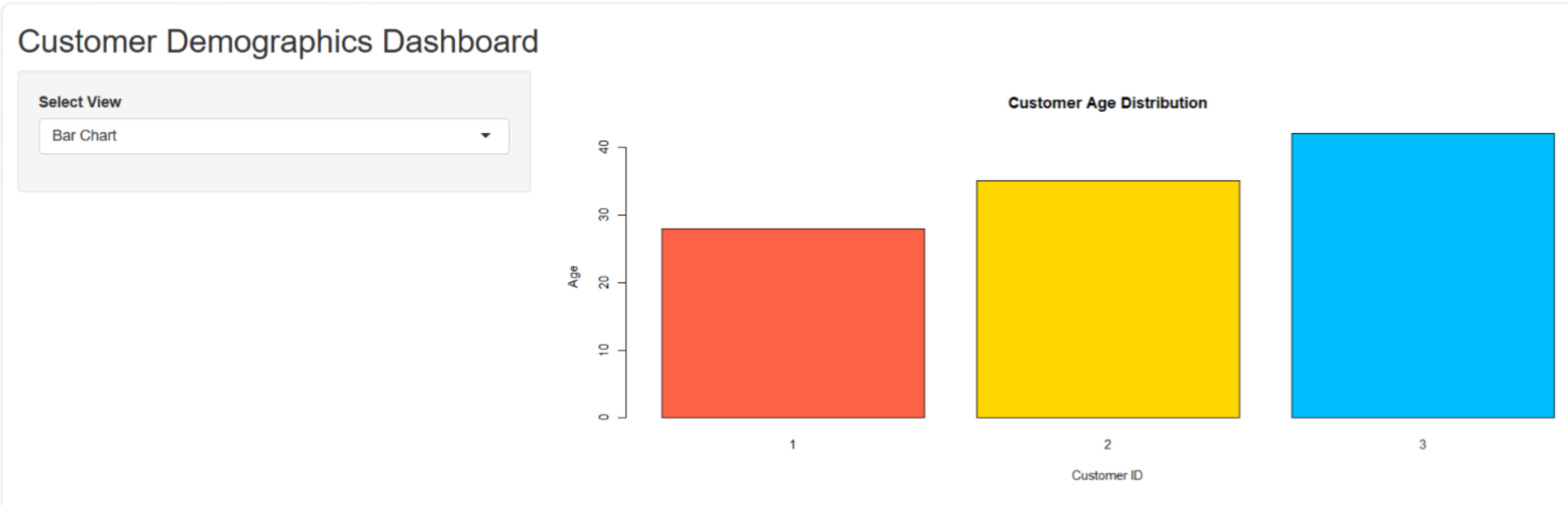
```
>
> print(customer)
  CustomerID Age Gender Income AgeGroup
1          1  28 Female  50000    20-30
2          2  35   Male  60000    31-40
3          3  42 Female  75000    41-50
>
```

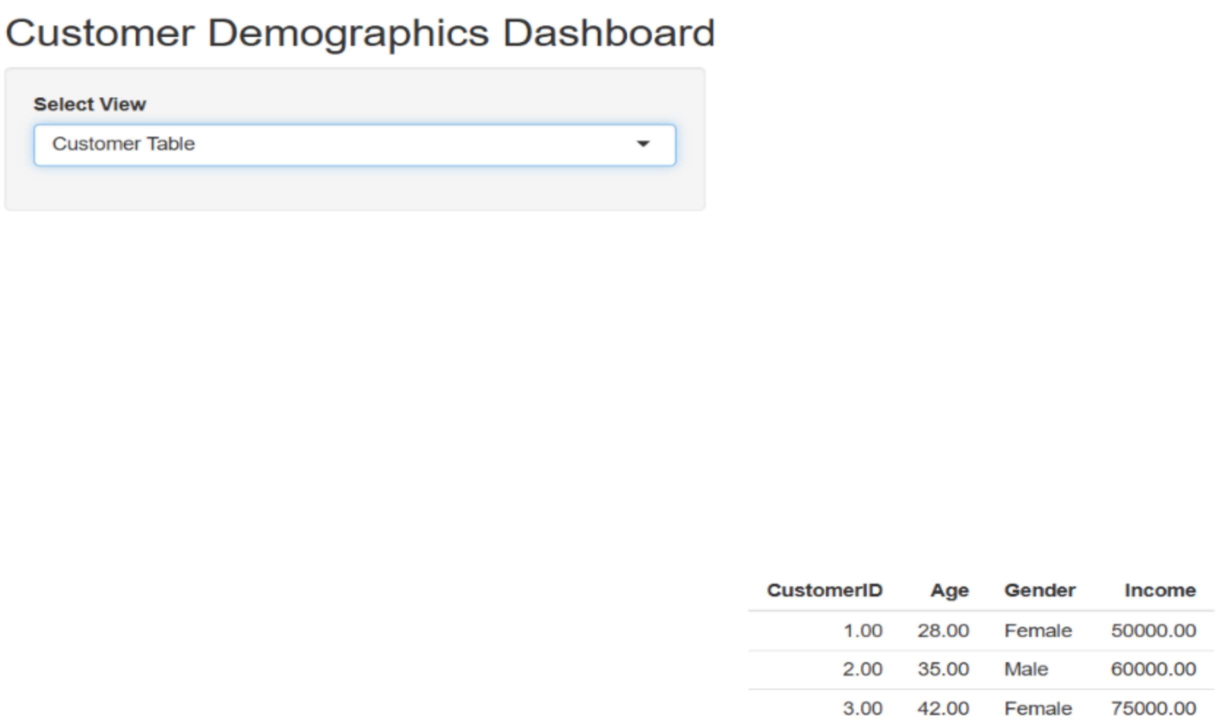
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
library(shiny)


customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ui <- fluidPage(

  titlePanel("Customer Demographics Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("view",
                  "Select View",
                  choices = c("Bar Chart",
                              "Pie Chart",
                              "Customer Table"))

    ),

    mainPanel(

      plotOutput("plot"),

      tableOutput("table")

    )

  )

)
```

```r
server <- function(input, output){

  output$plot <- renderPlot({

    if(input$view == "Bar Chart"){

      barplot(customer$Age,
              names.arg = customer$CustomerID,
              col = c("tomato","gold","deepskyblue"),
              border = "black",
              xlab = "Customer ID",
              ylab = "Age",
              main = "Customer Age Distribution")

    }

    else if(input$view == "Pie Chart"){

      gender <- table(customer$Gender)

      pie(gender,
          col = c("hotpink","skyblue"),
          main = "Customer Gender Distribution")

      legend("topright",
             legend = names(gender),
             fill = c("hotpink","skyblue"))

    }

  })

  output$table <- renderTable({

    if(input$view == "Customer Table"){

      customer

    }

  })
```

```
}

shinyApp(ui = ui, server = server)
```





CustomerID	Age	Gender	Income
1.00	28.00	Female	50000.00
2.00	35.00	Male	60000.00
3.00	42.00	Female	75000.00